Queen Mary
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EBU5376
Digital Signal Processing

DSP Systems – Frequency-domain Responses

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Outline

- Frequency Response
  - Amplitude Response
  - Phase Response
- Transfer Function

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  - Phase Response
- Transfer Function

Frequency Response

- The frequency response of a LTI system describes how the system responds to sinusoidal input signals of different frequencies.

- Specifically, it describes how the system modifies the amplitude and phase of the input signal at each frequency

Frequency Response

- The frequency response of a LTI system is denoted by $H(\omega)$ where $x[n] = e^{j\omega n}$ denotes the complex exponential input signal
- The system preserves the input frequency and modifies only its amplitude and phase, as determined by $H(\omega)$.

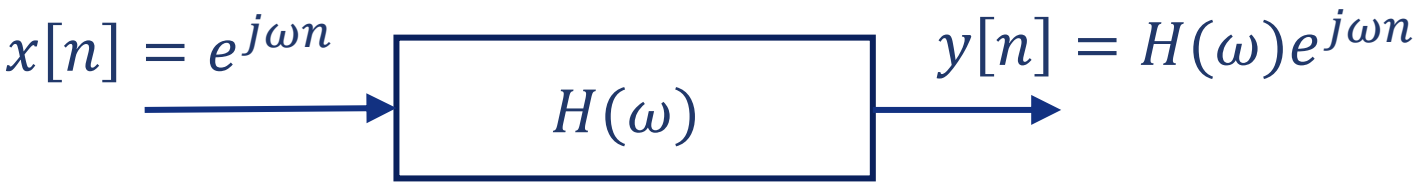

Fig: Representation of a LTI system using the frequency response.

From $h[n]$ to $H(\omega)$

- Recall that a LTI system based on impulse response is given by

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

- Now, suppose the input is a complex exponential function $x[n] = e^{j\omega n}$
- The frequency response is computed as

$$y[n] = \sum_{k=-\infty}^{\infty} h[n-k]e^{j\omega k} = \cdots = e^{j\omega n} \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$$

- Hence, we define the **frequency response** as

$$H(\omega) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$$

What is the name of this transformation?

Frequency Response - DTFT

- The **frequency response** can be obtained by applying the Discrete time Fourier transform (DTFT) to the **impulse response**, formulated as

$$H(\omega) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}$$

Discrete $\quad h[n] \xleftrightarrow{DTFT} H(\omega) \quad$ Continuous

Frequency Response – z-Transform

- DTFT is z-transform with $|z| = 1$
- The frequency response can also be obtained by applying the z-transform ($|z| = 1$) to the impulse response, given by

$$H(\omega) = \sum_{n=-\infty}^{\infty} h[n]z^{-n}, \qquad |z| = 1$$

Frequency Response

- Example: Derive the frequency response of

$$y[n] = 0.5x[n] + 0.25x[n-1]$$

Let $x[n] = \delta[n]$, the impulse response is $h[n] = 0.5\delta[n] + 0.25\delta[n-1]$
By applying DTFT to $h[n]$, we get

$$H(\omega) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} = h[0]e^{-j\omega 0} + h[1]e^{-j\omega}$$

$$= h[0] + h[1]e^{-j\omega}$$

$$= 0.5 + 0.25e^{-j\omega}$$

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Amplitude Response

- In the frequency domain, each point is characterized by its magnitude and phase, hence the frequency response can be written as

$$H(\omega) = |H(\omega)|e^{j\angle H(\omega)}$$

- The complex-valued frequency response may be decomposed into two <u>real-valued</u> functions:
  1. The amplitude response

$$|H(\omega)| = \sqrt{Re\big(H(\omega)\big)^2 + Im\big(H(\omega)\big)^2} = \sqrt{H(\omega)H^*(\omega)}$$

Amplitude Response

- Example: Derive the amplitude response of

$$H(\omega) = 0.5 + 0.25e^{-j\omega}$$

The amplitude response is computed by

$$|H(\omega)| = \sqrt{H(\omega)H^*(\omega)} = \sqrt{\left(\frac{1}{2} + \frac{1}{4}e^{-j\omega}\right)\left(\frac{1}{2} + \frac{1}{4}e^{j\omega}\right)} = \sqrt{\frac{5}{16} + \frac{1}{4}\cos\omega}$$

Amplitude Response

- Given that the amplitude response of a system is

$$|H(\omega)| = \sqrt{\frac{5}{16} + \frac{1}{4}\cos\omega}$$

- Can you describe the frequency domain behaviour of the system, based on this amplitude response?

Amplitude Response

- Given that the amplitude response of a system is

$$|H(\omega)| = \sqrt{\frac{5}{16} + \frac{1}{4}\cos\omega}$$

- Can you describe the frequency domain behaviour of the system, based on this amplitude response?

For discrete real-valued systems, the frequency ω is defined in the range $\omega \in [0, \pi]$

As the frequency increases from 0 to π, the value of $\cos\omega$ decreases from 1 to -1.

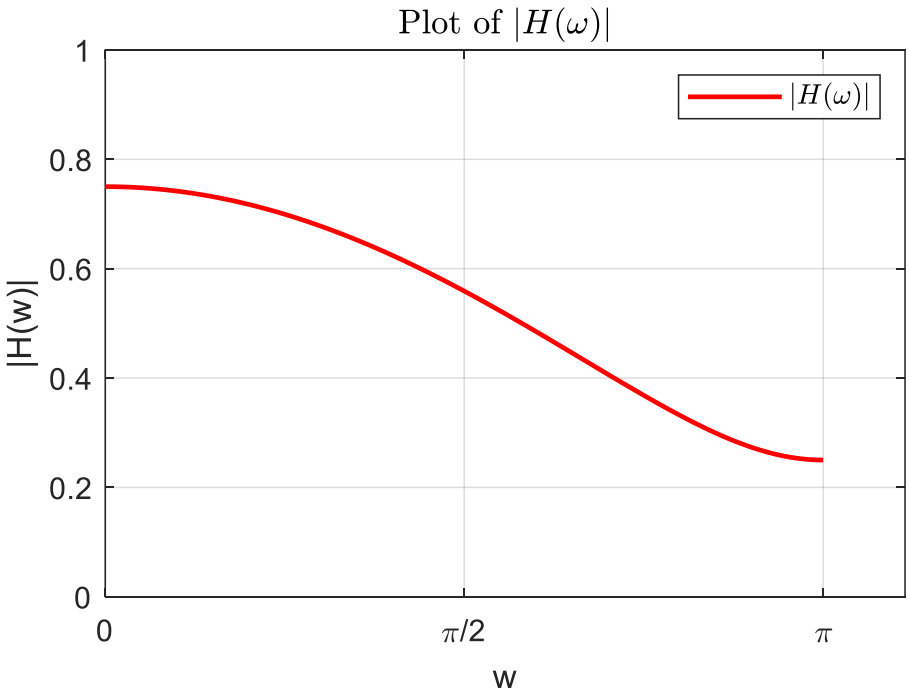
The maximum amplitude response is $|H(\omega)|_{max} \approx 0.75$ at $\omega = 0$

The minimum amplitude response is $|H(\omega)|_{min} \approx 0.25$ at $\omega = \pi$

The shape of the amplitude response is determined by the $\cos\omega$ term, which introduces a sinusoidal variation in the amplitude response.

Amplitude Response

- The amplitude response $|H(\omega)| = \sqrt{\frac{5}{16} + \frac{1}{4}\cos\omega}$ is plotted as



Frequency Response – Normalized Frequency

What is the meaning of ω in the frequency response?

- In analog systems, the frequency response describes how the system modifies the amplitude and the phase of the sinusoid input $e^{j\omega_a t}$
- For discrete systems, the sinusoid input is sampled as $e^{j\omega n}$:

$$e^{j\omega_a t} \rightarrow e^{j\omega n}$$

where $\omega = 2\pi \frac{f}{f_s}$ or equivalently $\omega = \frac{\omega_a}{f_s}$ is the **normalized frequency**

f denotes the signal frequency and f_s denotes the sampling frequency

- The concept of normalized frequency was introduced at the start of the discrete-time transformation chapter

Frequency Response – Normalized Frequency

What is the meaning of ω in the frequency response?

- From the previous slide, we know $\omega = 2\pi \frac{f}{f_s}$ is the **normalized frequency**
- To prevent aliasing, the sampling frequency must be more than twice the signal frequency, i.e., $f_s > 2f$
- When analyzing a discrete DSP system, we always assume that the system has an appropriate sampling rate, i.e., no aliasing
- Hence, when explaining/plotting the frequency response, the range for ω is typically $[0, \pi]$

Outline

- Frequency Response
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Phase Response

- The **phase response** determine the amount of time delay that the system introduces to the different frequency components of the input signal.

- The frequency response can be written as

$$H(\omega) = |H(\omega)|e^{j\angle H(\omega)}$$

The phase response is defined as

$$\angle H(\omega) = \tan^{-1}\left(\frac{Im\big(H(\omega)\big)}{Re\big(H(\omega)\big)}\right)$$

Phase Response

- Example: Suppose a system has the following frequency response:

$$H(\omega) = e^{-j\omega n_0}$$

The phase response is

$$\angle H(\omega) = -\omega n_0$$

Q: What impact does it have on the output signal?

Phase Response

- The phase response of $H(\omega) = e^{-j\omega n_0}$ is $\angle H(\omega) = -\omega n_0$

It implies that all frequency components in an input signal, will have the same phase delay of n_0 after passing through this system

To see this, consider the input $x[n] = e^{j\omega n}$, the corresponding output signal is computed as

$$y[n] = H(\omega)x[n] = e^{-j\omega n_0} \cdot e^{j\omega n} = e^{j\omega(n-n_0)}$$

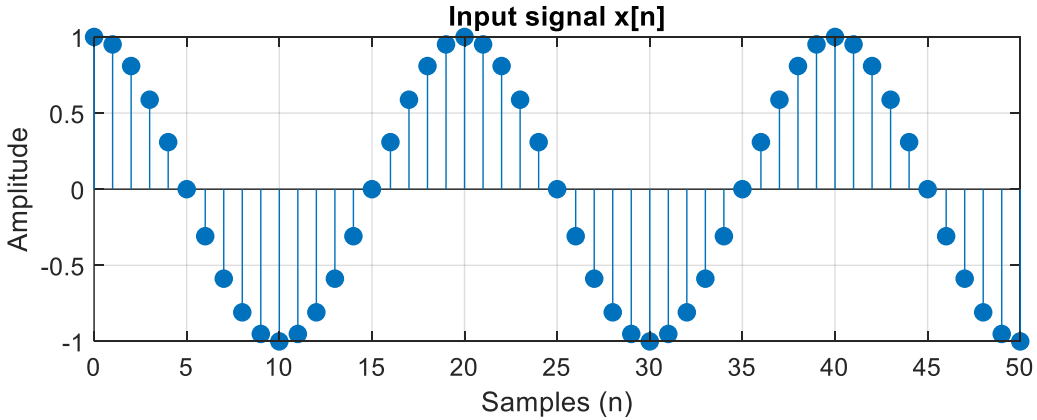
Hence, the output $y[n]$ is equivalent to the input $x[n - n_0]$, delayed by n_0 samples

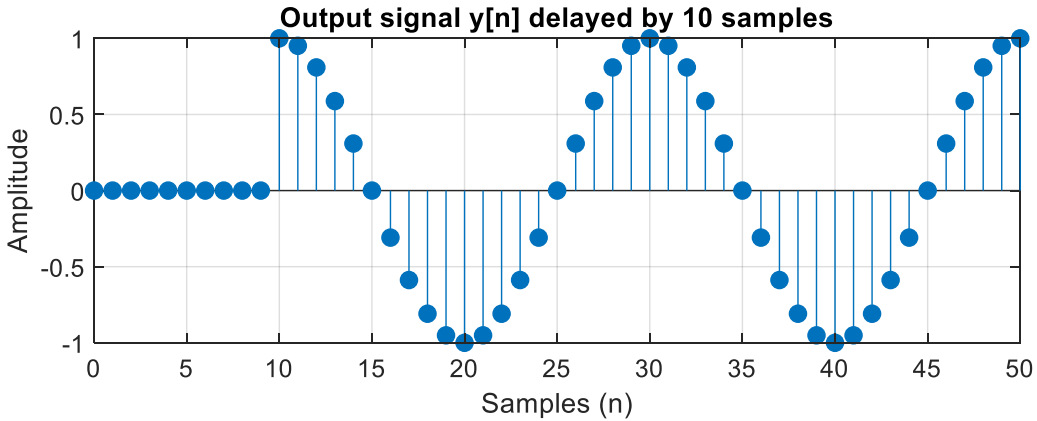
Phase Response

- Here is an example to visualize the phase delay
- The **input** is a 5 Hz cosine waveform sampled at 100 Hz:

  $$x[n] = \cos(\omega n), n = 0,1,2 \ldots$$

  where $\omega = 2\pi \frac{f}{f_s} = 0.1\pi$
- The system has a phase response of $\angle H(\omega) = -10\omega$, i.e. $n_0 = 10$
- The **output** is $y[n] = \cos\big(\omega(n-10)\big), n = 10, 11, \ldots$





Phase Response

- Example: Suppose a system has the following frequency response:

$$H(\omega) = e^{-j\omega^2}$$

The phase response is

$$\angle H(\omega) = -\omega^2$$

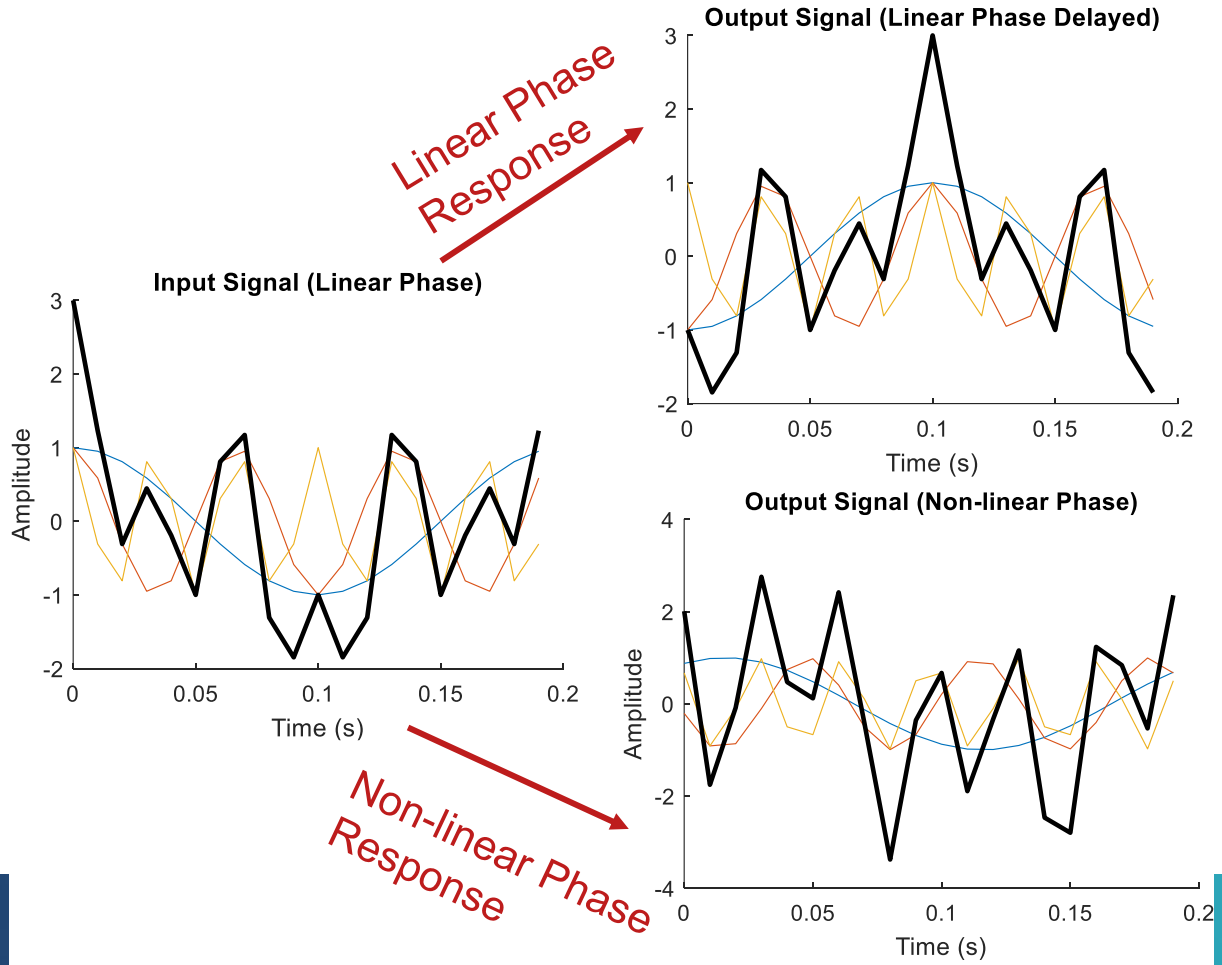
It implies that the phase delay increases quadratically as the (normalized) frequency ω increases.

This is a non-linear phase response.

In a composite signal, higher frequency components suffer from a greater phase delay compared to the lower frequency components. As a result, the shape of the composite signal will change.

Phase Response

- These figures illustrate the impact of **linear** and **non-linear phase response** on the same input signal
- The black line is the signal, while the colored lines are the component frequencies



Phase Response - Linearity

- The phase response is always a function of the (normalized) frequency ω
  - If this function is a linear function, we say the phase response is linear, e.g. $\angle H(\omega) = -\omega n_0$
  - If this function is a non-linear function, we say the phase response is non-linear, e.g. $\angle H(\omega) = -\omega^2$
- If a DSP system has a linear phase response, it means all frequency components in the input are shifted/delayed by the same amount. Hence, the shape of the signal is unchanged
- Otherwise, the system will change the shape of the signal

Outline

- Frequency Response
  - Amplitude Response
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- Transfer Function

Transfer Function – Discrete LTI System

- Recall, the LTI system output can be written as the discrete convolution between its input and the impulse response

$$y[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

- The convolution property of z-transform states that, convolution in time-domain is multiplication in z-domain. Hence,

$$Z(y[n]) = Z(h[n])Z(x[n])$$

$$\Rightarrow Z(h[n]) = \frac{Y(z)}{X(z)}$$

Transfer Function – Discrete LTI System

- For discrete LTI systems, transfer function is defined as the ratio between the system output and input in the z-domain

$$H(z) = \frac{Z(y[n])}{Z(x[n])} = \frac{Y(z)}{X(z)}$$

- Equivalently, the transfer function can be obtained as the z-transform of the impulse response, i.e.,

$$H(z) = Z(h[n])$$

Transfer Function – Continuous LTI System

- For continuous LTI systems, transfer function is defined as the ratio of the Laplace transform of the system's output to the Laplace transform of the system's input

$$H(s) = \frac{\mathcal{L}\big(y(t)\big)}{\mathcal{L}\big(x(t)\big)} = \frac{Y(s)}{X(s)}$$

Transfer Function – Relation to Difference Equation

Relationship between Difference Equation and Transfer Function:

Difference Equation: $$y[n] = -\sum_{i=1}^{N} a_i y[n-i] + \sum_{j=0}^{M} b_j x[n-j]$$

Transfer Function: $$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

Transfer Function – Relation to Frequency Response

- We now know that
  - ✓ Frequency response is DTFT of impulse response
  - ✓ Transfer function is z-transform of impulse response

- Recall that DTFT is a special case of z-transform with $|z| = 1$

- Hence frequency response is the transfer function evaluated on the unit circle

Transfer Function

- Example 1: Derive the transfer function of

$$y[n] = 0.5x[n] + 0.25x[n-1]$$

Let $x[n] = \delta[n]$, the impulse response is $h[n] = 0.5\delta[n] + 0.25\delta[n-1]$
By applying z-transform to $h[n]$, we get

$$H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n} = h[0]z^{-0} + h[1]z^{-1}$$

$$= 0.5 + 0.25z^{-1}$$

Transfer Function

- Example 2: Derive the difference equation of the system based on $H(z) = 0.5 + 0.25z^{-1}$

We need to express the transfer function in the form

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

In fact,

$$H(z) = \frac{0.5 + 0.25z^{-1}}{1} \Rightarrow a_0 = 1, b_0 = 0.5, b_1 = 0.25$$

Hence, the difference equation is $y[n] = 0.5x[n] + 0.25x[n-1]$

Transfer Function

- Example 3: Derive the difference equation from

$$H(z) = \frac{(z+1)^2}{\left(z - \frac{1}{2}\right)\left(z + \frac{3}{4}\right)}$$

We can expand the fractions to

$$H(z) = \frac{z^2 + 2z + 1}{z^2 + \frac{1}{4}z - \frac{3}{8}} = \frac{1 + 2z^{-1} + z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}}$$

Assuming the system is causal,

$$y[n] + \frac{1}{4}y[n-1] - \frac{3}{8}y[n-2] = x[n] + 2x[n-1] + x[n-2]$$

Summary

- **Transfer function** $H(z)$ is the frequency-domain representation of a LTI system
  - Difference equation is the time-domain representation
  - Derived from the z-transform of the difference equation
  - Can also be derived as the z-transform of the impulse response

$$H(z) = \frac{Z(y[n])}{Z(x[n])} \qquad H(z) = Z(h[n])$$

Summary

- **Frequency response** $H(\omega)$ describes how the system reacts to a sinusoid input of unit amplitude, i.e., $e^{j\omega}$
  - Can be derived from the transfer function $H(z)$ by substituting $z = e^{j\omega}$
  - Can be derived as the DTFT of the impulse response
- Frequency response is complex-valued, which has an amplitude and a phase.

Summary

- When evaluating the frequency-domain characteristics of a system, we often look at two separate responses:
  1. **Amplitude response**
     - How system changes the amplitude of a sinusoid input
  2. **Phase response**
     - How system changes the phase of a sinusoid input

Summary

- The **transfer function** carries another important information of the LTI system – Stability
- Stable systems are easier to control, whereas unstable systems can lead to disasters, e.g., outputs with infinite amplitude
- The stability of a system can be analysed by looking at the poles of the transfer function.
- This is the topic of the next lecture

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